Binary Tree Node Counting Using C++

Problem Statement

Write a C++ program to create a binary tree and count the total number of nodes, leaf nodes, internal nodes, and nodes having exactly one child. The program should use a recursive function to traverse the binary tree and identify each type of node based on its left and right child pointers.

A leaf node is a node that has no children, while an internal node is a node that has at least one child. A node having exactly one child is a node with either a left child or a right child, but not both.

The program should initialize the required counters, traverse the entire binary tree recursively, classify each node, and update the corresponding counters. Finally, it should display the total number of nodes, leaf nodes, internal nodes, and nodes having exactly one child.

Objectives

- To understand the concept of a binary tree in data structures.
- To create a binary tree using C++ programming.
- To traverse a binary tree using recursion.
- To identify and count leaf nodes.
- To identify and count internal nodes.
- To count nodes having exactly one child.
- To display the total count of each type of node.

Expected Output

The program should display:

- Total number of nodes.
- Total number of leaf nodes.
- Total number of internal nodes.
- Total number of nodes having exactly one child.